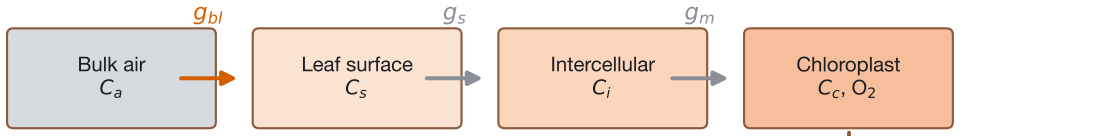
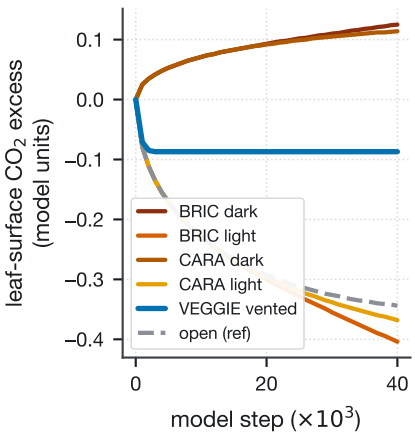
# Growth hardware, not gravity, sets the atmosphere at the leaf

## a The conductance chain

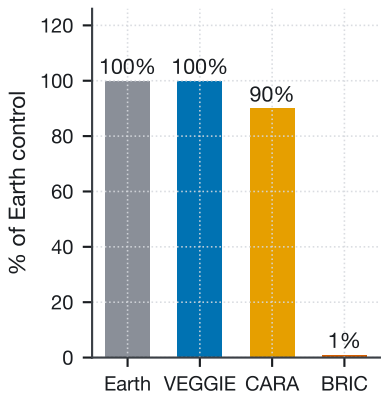



A sealed, lit enclosure lowers $C_a$ itself — upstream of every conductance in the chain.

## b Leaf-surface $CO_2$, microgravity




## c Carbon gained over 12 h




## d How fast a sealed canister changes

BRIC lit: $CO_2$ 400 → ~0 ppm
**7 minutes**

BRIC dark: $CO_2$ → 1%
**9.4 hours**

BRIC dark: $O_2$ 21% → 5%
**6.5 days**

CARA / VEGGIE: vents to cabin
**bounded**

The 7-minute drawdown is a mass balance, so it happens in the ground control too — which is why it cancels out of flight-vs-ground.

Gas transport from LunarLeaf-CFD (validated D2Q9 LBM solver); all hardware runs are in microgravity.